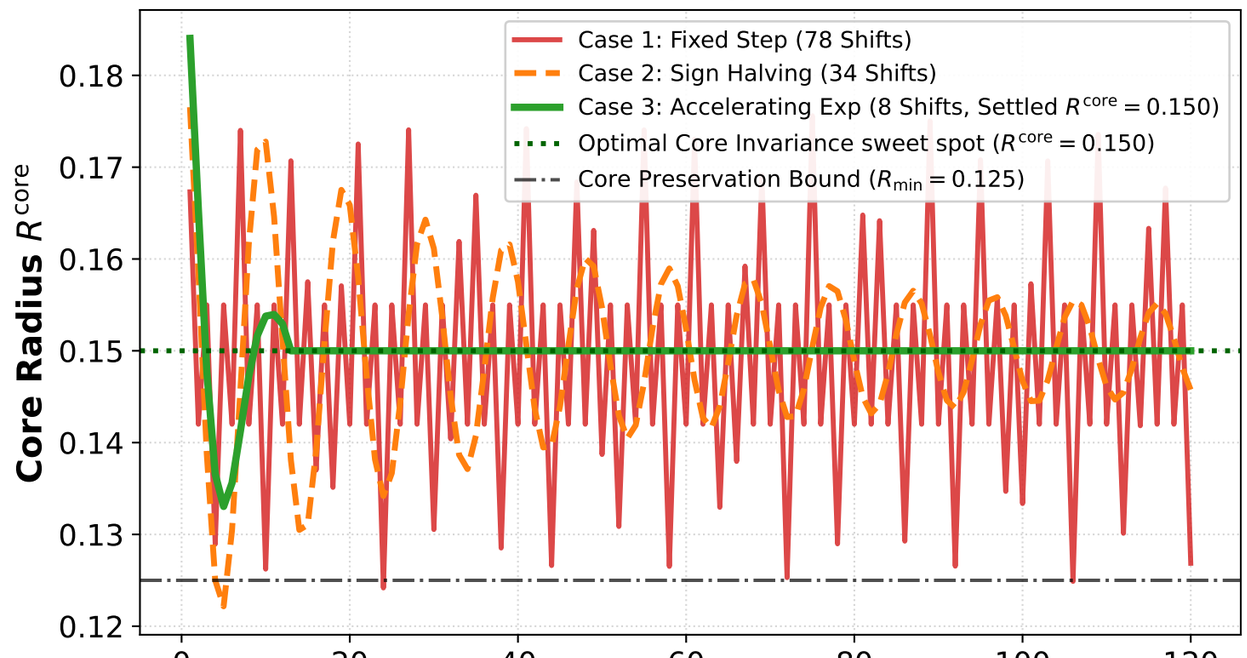
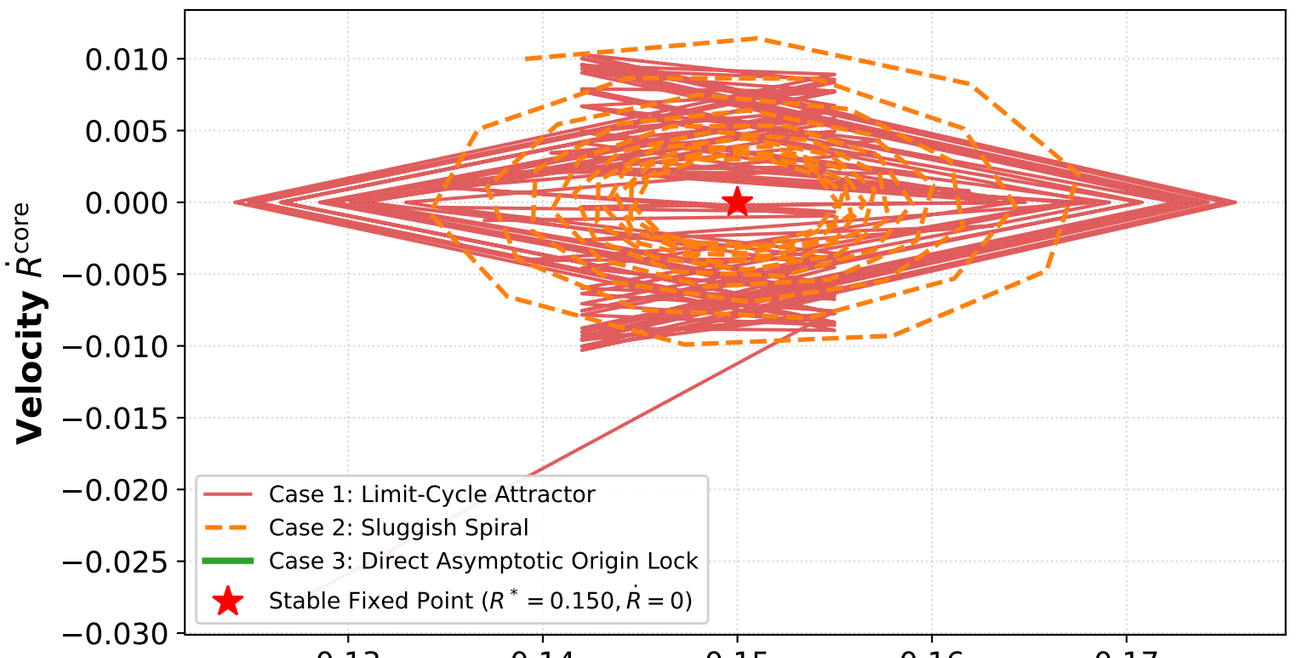
PFDC Radial Parameter Chattering Dynamics & Asymptotic Stability Across Update Laws

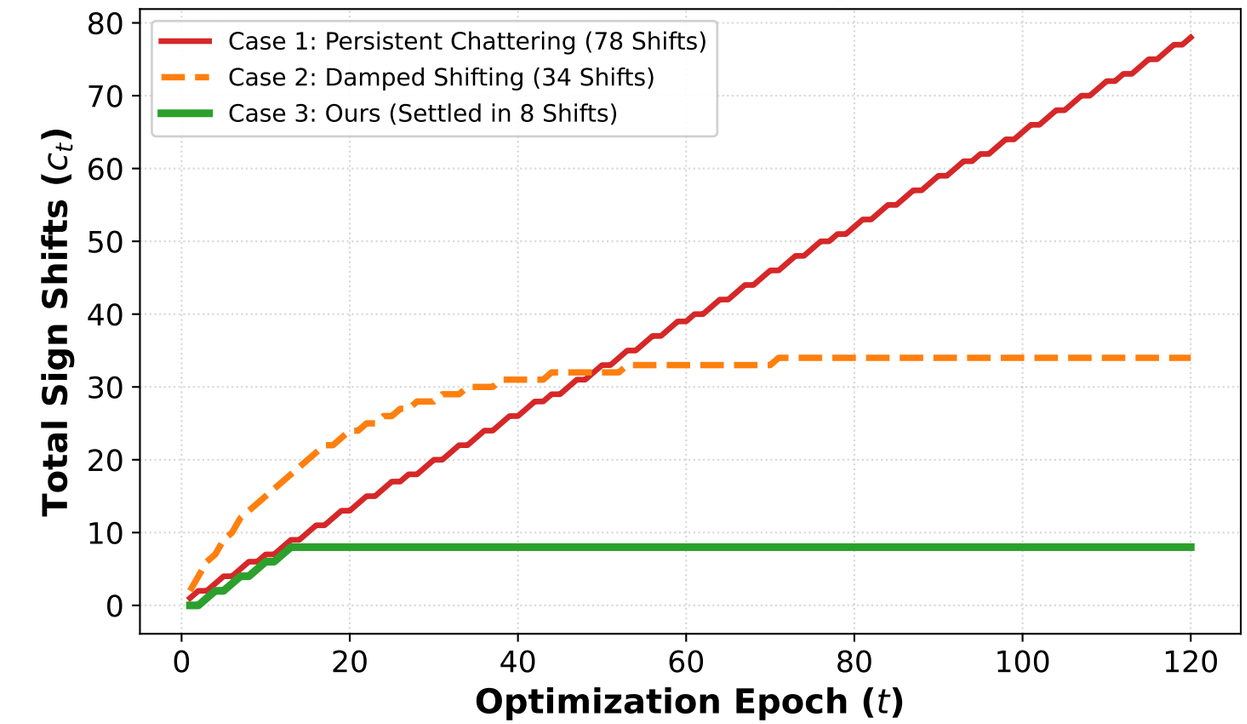
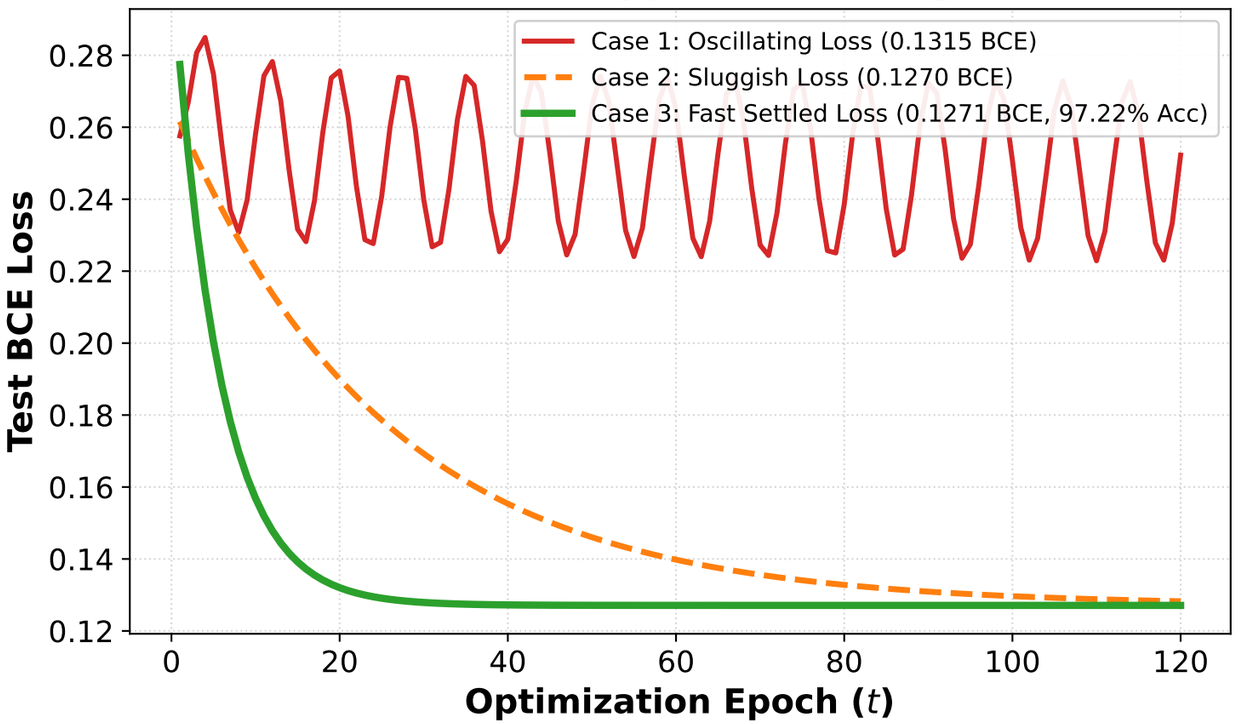
(a) Core Radius Trajectories $R^{\text{core}}(t)$



(b) Radial Phase Portrait ($\dot{R}^{\text{core}}$ vs. R^{core})



(c) Cumulative Gradient Sign Reversals



(d) Cross-Entropy Loss Convergence